



JEPPIAAR
ENGINEERING COLLEGE

DEPARTMENT OF COMPUTER SCIENCE ENGINEERING

CS3811- Project Work

Domain: Artificial Intelligence

AI-Driven Frustration Detection for Smarter Code Reviews

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Abstract

- **The Problem:** Code reviews are mentally draining, causing reviewer burnout and project delays. Existing tools only analyze code, not the reviewer's state.
- **Our Solution:** AFFECT-CODE is an intelligent assistant that detects reviewer frustration in real-time and provides subtle help to improve focus and feedback quality.
- **The Outcome:** A smoother, kinder review process that enhances developer well-being and team productivity through human-aware AI.

Existing System

Current Tools Are "Code-Centric":

- Automated linters and AI assistants (e.g., GitHub Copilot) focus only on finding bugs in the code.
- They completely ignore the reviewer's cognitive load, frustration, or focus levels during the process.

The Research Gap We Identified:

- Studies confirm keystroke/mouse patterns can reveal a developer's emotional state (frustration, flow).
- Other research highlights frustration as a major bottleneck in code reviews.
- No tool exists that connects these two insights to assist the reviewer in real-time. Our Idea bridges this gap.

Advantages of Proposed System

For the Individual Reviewer:

- Reduces mental fatigue and frustration during difficult reviews.
- Provides timely, non-intrusive prompts to maintain focus and prevent burnout.

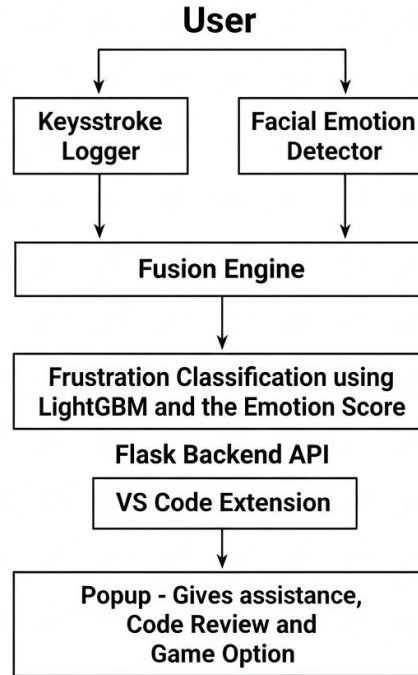
For the Development Team:

- Improves the quality and constructiveness of peer feedback.
- Gives team leads anonymized analytics on review sentiment to better manage workflow.

For the Software Process:

- Introduces a novel "reviewer well-being" metric into development analytics.
- Potential to accelerate review cycles by reducing friction and mental blockers.

System Architecture



SOFTWARE AND HARDWARE REQUIREMENTS

Software Stack:

- Development: VS Code, Node.js, Python, Git.
- Core Tech: React.js, , Flask, scikit-learn, TensorFlow,Keras,OpenCV,Deepface,.

Hardware Requirements:

- Standard developer laptops are sufficient.
- Minimum: 8GB RAM, Modern CPU (Intel i5 / Ryzen 5 or equivalent).
- Note: No high-end GPU is needed as our ML model uses efficient CPU-based algorithms.

List Of Modules

- Data Collection & Labeling Module
- Feature Engineering & Model Training Module
- Backend API Development Module
- VS Code Extension Development Module
- Frontend Dashboard Development Module
- System Integration & Testing Module
- Deployment & Packaging Module

Implementation Plan

- Phase 1 : Research & AI Model Prototype. Finalize the core frustration detection logic.
- Phase 2 : Core Development. Build the VS Code extension and backend APIs.
- Phase 3 : System Integration. Connect all components and implement basic interventions.
- Phase 4 : Frontend & Testing. Develop the dashboard and conduct initial user tests.
- Phase 5 : Finalization & Paper Draft. Deploy the system, create a demo, and draft the research paper.